

**28 June 2026**

**COS1501**

**Theoretical Computer Science**

25 marks

Duration : 1 Hours

Welcome to the COS1501 Assignment 3.

Primary Lecture name: Thapelo Andrew Sindane

Internal moderator name: Dr Petra Le Roux

This paper consists of 3 pages.

Instructions:

- Please note that NO other form of assistance is allowed (another individual, notes, internet or anything else).
- Cutting and pasting from the textbook (electronic media) is prohibited.
- You have 1 hours to complete this paper.
- You may upload PDF files only, nothing else is acceptable.
- Be sure that your upload is NOT password protected or uploaded as read only.
- Answer all questions and in sequence.

- Incorrect file format and uncollated answer scripts will not be considered.
- NO emailed scripts will be accepted.
- Students are advised to preview submissions (answer scripts) to ensure legibility and that the correct answer script file has been uploaded.
- Incorrect answer scripts and/or submissions made on unofficial assessment platforms (including the invigilator cell phone application) will not be marked and no opportunity will be granted for resubmission.
- Submissions will only be accepted from registered student accounts.
- Students suspected of dishonest conduct during the assessment will be subjected to disciplinary processes. UNISA has a zero tolerance for plagiarism and/or any other forms of academic dishonesty.

### Question 1 [Theory and Understanding]

1.1 Explain the following concepts in your own words. Your explanations should reflect the underlying mathematical principles and use appropriate notation where necessary.

- |                         |     |
|-------------------------|-----|
| a. Set Relation         | (1) |
| b. Function             | (1) |
| c. Reductio ad absurdum | (1) |
| d. Domain               | (1) |
| e. Trichotomy           | (1) |

Sub Total [5]

### Question 2 [Concept and application]

2.1 Let  $Q = \{(1,a), (3,b), 5, (7,c), (9,d), 11\}$  and  $T = \{2,4\}$  be sets. Find the Cartesian product of the following  $Q \times T$ , and describe briefly what your steps (2)

2.2 Let  $S = \{(e,e), (e,p), (p,e), (q,r), (p,s)\}$  be a relation on the set  $H = \{e,p,q,r,s\}$ .

- |                                                                                                                                          |     |
|------------------------------------------------------------------------------------------------------------------------------------------|-----|
| a. Examine the relation $S$ and state with justification, whether it is reflexive, irreflexive, symmetric, antisymmetric, and transitive | (5) |
|------------------------------------------------------------------------------------------------------------------------------------------|-----|

2.3 Let  $P$  be defined as the inverse relation of  $S$ .

- |                                             |     |
|---------------------------------------------|-----|
| a. Determine the relation composition $P;S$ | (3) |
|---------------------------------------------|-----|

Sub Total [10]

Question 3 [Complex Theorems and Problem Solving]

- 3.1 Proof that if  $R$  is a relation on a set of Positive Integers  $Z$ , then  $R$  is symmetric if and only if  $R$  is the same as its inverse. (2)
- 3.2 Let  $W = \{1,2,3\}$  be a set, and  $G = \{\{1\}, \{2,3\}\}$  be defined from  $W$ . Determine if  $G$  is a partition from  $W$ , and if it is, describe it's equivalence relation. (3)
- 3.3 Let  $D = \{(x,y) \mid y = x+1\}$  be a relation from  $X$  to  $Y$ , where  $X=Y=Z$ . Determine if  $D$  is a function (2)
- 3.4 Let  $f:Z \rightarrow Z$  be a function defined by  $f(x) = 3-x$ . Check if the function comply with the property  $f$  injectivity, surjectivity, and bijectivity (3)

Sub Total [10]

**TOTAL [25]**